Fiscal Year:	FY 2011	**Task Last Updated:**	FY 06/23/2011
PI Name:	Matson, Douglas Ph.D.		
Project Title:	Electromagnetic Levitation Flight Support for Transient Observation of Nucleation Events (ELFSTONE)		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	douglas.matson@tufts.edu	**Fax:**	FY (617) 627-3058
PI Organization Type:	UNIVERSITY	**Phone:**	(617) 627-3239
Organization Name:	Tufts University		
PI Address 1:	Mechanical Engineering		
PI Address 2:	025 Anderson Hall, 200 College Ave		
PI Web Page:	https://		
City:	Medford	**State:**	MA
Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	09/20/2010	**End Date:**	09/19/2013
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	Douglas Matson is U.S. Team Member on ESA-sponsored projects, "Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC)" and "Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS)."		
COI Name (Institution):			
Grant/Contract No.:	NNX10AV27G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Funding is for Dr. Matson's role as Team Member on the subprojects described below, associated with ESA-sponsored main projects as noted. Main project: Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC); Science Coordinator is Dr. Thomas Volkmann, German Aerospace Center DLR, Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany Subproject: Transformation kinetics - ISS Using results from thermophysical property evaluation and combining these with the MHD convection definition, we will set convection over as wide a range as is possible and conduct recalescence studies from the undercooled melt using the MSL-EML to compare to results from ground-based EML and ESL. The critical parameters of interest are evaluation of the incubation time between formation of metastable phases and subsequent transformation to the stable phase for Fe-based and titanium aluminide alloys. Since this transformation is very rapid, the high-speed video system must be used to image the processes associated with nucleation and growth during phase selection. Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany Subprojects: Management and Coordination. Coordination of the scientific program, Coordination of the ground based measurement program, Coordination of the microgravity experiment program, Contact with the national and international space agencies. Short Duration Microgravity Measurements - PF. Conduct parabolic flights on selected alloys. Long Duration microgravity Experiments - ISS. Surface tension and viscosity as a function of temperature, Density as a function of temperature, Solidification growth velocity. Preparation of the Microgravity Experiments. Experimental protocols for processing on the ISS , Verification of flight samples for the ISS. MODELLING A: Thermodynamics, Phase Selection. Thermodynamic phase diagram calculation, Modeling of the influence of convection on phase selection.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The value of this research is that the results have application to the design of industrial welding, spray forming and casting operations for a commercially important class of soft magnetic materials. In addition, this research addresses fundamental issues relating to rapid solidification behavior, metastable phase selection and analysis of the processes governing microstructural evolution. Experiments focus on the use of convection as a processing control parameter.
Task Progress:	Highlights of Work Accomplished During Grant Period September 2010-11. • Invited member ESA International Topical Team on Thermophysical Properties (ThermoLab-ISS kick-off meeting) at ESA/ESTEC 11/21/2010 • Conducted ESL viscosity and density tests on FeCo alloys at MSFC 1/10-16/2011 • Conducted sounding rocket microgravity solidification experiments 28 March 2011 at Esrange Space Center, Kiruna SE in support of PARSEC project • Conducted ESL viscosity tests at ESL-MSFC 7-16 June 2011 • Hired Tufts MS student Jackson Dolan to conduct GB FeCo testing • Hired Tufts undergraduate student Robert Hayes to conduct flow modelling • Invited talk at TU-Graz for Prof. Gernot Pottlacher on microgravity testing • Invited talk at International Workshop on Dynamics, Properties and Solidification of Liquid Alloys Symposium in Koln DE 7-9 July 2011 • Delivered presentation at ISPS conference in Bad Godesburg DE 11-15 July 2011 • Attended IWG-4 at ESA HQ in Noordwijk ND 18-19 July 2011
Bibliography Type:	Description: (Last Updated: 05/05/2026)
Books/Book Chapters	Matson DM. "Chapter 11 - Nucleation within the Mushy Zone." in "Solidification of containerless undercooled melts." Ed. D.M. Herlach, D.M. Matson. Wiley-VCH, in press, June 2011. Expected publication Fall 2011., Jun-2011
Papers from Meeting Proceedings	Matson DM, Hyers RW. "Phase selection in the Mushy Zone: LODESTARS and ELFSTONE Programs." 4th International Symposium on Physical Sciences in Space, Bonn, Bad Godesberg, Germany,11-15 July 2011. Proceedings. 4th International Symposium on Physical Sciences in Space, Bonn, Bad Godesberg, Germany,11-15 July 2011. p. 220-225. , Jul-2011